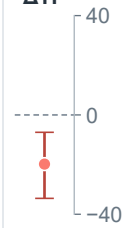
IEEE-CIS | Combined, component and workflow effects

Exact 3% completed-Test-window budget · raw scores · fixed workflows

a Fold 1 | Combined effect

LightGBM · M2 − M0

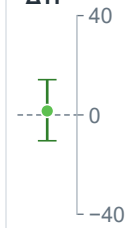
ΔTP	−19.8 [−33.6, −7.0]
ΔRecall (pp)	−0.946 [−1.594, −0.340]
ΔAP (0–1)	−0.0039 [−0.0084, 0.0003]



b Fold 2 | Combined effect

LightGBM · M2 − M0

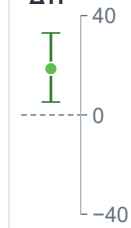
ΔTP	+1.6 [−10.4, 14.2]
ΔRecall (pp)	+0.078 [−0.507, 0.684]
ΔAP (0–1)	−0.0019 [−0.0059, 0.0022]



c Fold 3 | Combined effect

LightGBM · M2 − M0

ΔTP	+18.6 [5.2, 33.0]
ΔRecall (pp)	+0.840 [0.232, 1.522]
ΔAP (0–1)	+0.0188 [0.0135, 0.0241]



d Fold 1 | Component effects

Workflow	H	W
LightGBM	+3.4	−23.2
Logistic	−7.0	+11.0
Direct	−10.4	+34.2

Direct 95% intervals
H: [−23.8, 4.6]
W: [16.6, 50.6]

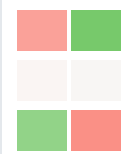
e Fold 2 | Component effects

Workflow	H	W
LightGBM	−3.4	+5.0
Logistic	+14.0	+1.0
Direct	+17.4	−4.0

Direct 95% intervals
H: [0.0, 35.6]
W: [−18.0, 9.2]

f Fold 3 | Component effects

Workflow	H	W
LightGBM	−81.6	+100.2
Logistic	−2.0	+1.0
Direct	+79.6	−99.2



Direct 95% intervals
H: [58.8, 98.0]
W: [−115.8, −71.6]

H: history (M1 − M0) W: weighting above history (M2 − M1)

Direct = Logistic effect − LightGBM effect; all component values are ΔTP.

Pointwise conditional 95% intervals; colours encode direction, not significance.

